

Who are the digitally excluded in Italian finance?

Rubini, Asia ID: 882663 a.rubini7@campus.unimib.it
Villa, Valeria ID: 895029 v.villa47@campus.unimib.it

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Abstract

This study investigates the determinants of digital financial exclusion among Italian adults using data from the 2023 IACOFI survey (Banca d'Italia, N = 4,862). Digital exclusion is operationalised at two levels — absence of internet access and low use of digital financial tools — yielding an overall exclusion rate of 43%. A weighted logistic regression identifies older age, retirement, low income, and Southern residence as independent risk factors, while higher financial literacy is protective. A k-means clustering on internet users reveals three distinct excluded profiles: the Doubly vulnerable (low literacy and low income), the Literate late adopters (behavioural barriers despite high knowledge), and the Educated but excluded elderly (72% exclusion rate). Results show that digital financial exclusion in Italy is heterogeneous and that financial education is an independent lever for inclusion, beyond socio-demographic factors alone.

Keywords: digital financial exclusion, financial literacy, logistic regression, k-means clustering, IACOFI 2023

Dataset: IACOFI 2023 — *Indagine sulle competenze e le abilità finanziarie dei cittadini italiani*, Banca d'Italia. Available at: <https://www.bancaditalia.it/statistiche/tematiche/indagini-famiglie-imprese/alfabetizzazione/>

AI tools declaration: AI tools (Claude, Gemini) were utilized strictly in a supportive capacity. Specifically, they were employed as technical aids for syntax verification and code troubleshooting during the implementation of the Python pipelines, as well as for language polishing and proofreading of the manuscript. The conceptualisation of the research design, the operationalisation of the variables, and all analytical interpretations remain entirely the exclusive work of the authors.

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1 Research Question

1.1 Statement

This study investigates the following research question:

What socio-demographic and financial literacy characteristics predict digital financial exclusion among Italian adults, and can identifiable vulnerability profiles be detected within the excluded population?

For the purposes of this study, *digital financial exclusion* is defined as the non-adoption of digital financial tools — that is, the absence or near-absence of use of online or mobile channels to perform financial activities such as checking account balances, executing payments, transferring funds, or accessing investment and credit services. The concept is operationalised at two distinct levels, reflecting the structure of the data:

- **Level 0 — Basic digital divide:** the respondent has no internet access at all (variable QD14 = 0) and is therefore excluded from every digital financial service by construction.
- **Level 1 — Digital financial exclusion conditional on internet access:** the respondent has internet access but makes low or no use of digital financial tools, as measured by a composite **Digital Adoption Score (DAS)** built from the QP8 block (five binary items on whether the respondent has ever performed a given financial operation fully online) and the QP9 block (seven items on the frequency of digital financial activities over the past twelve months), rescaled to a 0–100 range. Respondents at or below the weighted 33rd percentile of the DAS distribution among internet users are classified as “Internet, low digital use”.

The two levels together define a binary outcome variable (*digitally excluded* vs. *digital adopter*) used in the logistic regression, and a three-category variable used in the descriptive profiling and cluster interpretation.

Financial literacy is operationalised as an objective score (0–7), following the standard OECD/INFE methodology, built from seven knowledge questions covering inflation, simple and compound interest, risk-return trade-offs, diversification, and mortgage repayment (items QK3, QK5, QK6, QK7_1, QK7_2, QK7_3, QK10 of the IACOFI questionnaire). Each item is coded 1 if answered correctly and 0 otherwise, including “don’t know” responses.

Socio-demographic characteristics considered in the analysis are:

- Gender (QD1);
- Age (QD7/QD7_A);
- Macro-region of residence (QD2);

- Type of municipality (QD3) ;
- Education level (QD9, collapsed to three ordinal categories: Low, Medium, High)
- Employment status (QD10, collapsed to four groups: Employed, Retired, Not in labour force, Student)
- Household income bracket (QD13).

1.2 Motivation and stakeholders

The diffusion of digital financial services — online banking, mobile payments, digital investment platforms — has accelerated markedly across Italy and Europe in the past decade. While this shift offers significant benefits in terms of cost, convenience, and access to a broader range of financial products, it also creates a risk of *exclusion by design*: individuals who do not adopt digital financial tools may be progressively left behind as physical branches close, cash handling becomes less convenient, and digital-only products proliferate. Understanding *who* these individuals are, and *why* they are excluded, is therefore a policy question of growing urgency.

This study is of direct interest to several categories of stakeholders:

- **Banca d'Italia and national financial regulators**, who commission the IACOFI survey precisely to monitor the financial capability and inclusion of the Italian population, and who need evidence-based guidance on which segments of the population require targeted intervention [1].
- **Financial education programme designers**, at the national level (e.g. the *Comitato per la programmazione e il coordinamento delle attività di educazione finanziaria*, EDUFIN) and at the level of individual financial institutions. The identification of distinct excluded profiles — rather than a single homogeneous "non-adopter" category — is directly actionable: it suggests that a one-size-fits-all financial education approach is unlikely to be effective, and that different profiles require different types of intervention (e.g. digital literacy training for the basic-divide group, trust-building and behavioural nudges for the "literate late adopters" cluster).
- **Policy makers working on the digital transition** (e.g. the Italian *Piano Nazionale di Ripresa e Resilienza*, PNRR, which includes measures to reduce the digital divide), for whom evidence on the overlap between digital exclusion and financial vulnerability is relevant to prioritising resources and target populations.
- **Researchers** in the fields of household finance, financial literacy, and digital economics, for whom the IACOFI dataset offers a nationally representative snapshot of Italy in 2023 — the first edition of the survey to include a block of digital financial behaviour questions following the updated OECD/INFE 2022 framework [12].

The key value-added of this study, relative to a purely descriptive use of the survey, lies in combining multivariate modelling (which allows disentangling the independent contributions of socio-demographic factors and financial literacy) with unsupervised clustering (which allows identifying internally coherent *profiles* of excluded individuals rather than only ranking risk factors). The interaction between these two methods is designed to produce both statistically sound estimates and substantively interpretable results for non-technical stakeholders.

2 The Data

The dataset used in this study is the **IACOFI 2023** (*Indagine sulle competenze e le abilità finanziarie dei cittadini italiani*, Survey on the Financial Literacy and Financial Skills of Italian Citizens), collected by the **Banca d'Italia** and publicly available on its institutional website [1]. IACOFI is conducted periodically on a sample of the Italian adult population aged 18 to 79, using computer-assisted telephone interviews (CATI). The 2023 edition is the most recent available at the time of this study and is the first to include an extended set of questions on digital financial behaviour, explicitly aligned with the updated OECD/INFE 2022 measurement framework for financial literacy, making it the most suitable edition for the purposes of this research question.

2.1 Data description

The raw dataset contains 4,862 respondents and 219 variables, collected on a sample of the Italian adult population (18-79 years old). Each respondent is associated with a survey weight (wght), normalised so that the sum of the weights equals the sample size (4,862), which is used in all the descriptive statistics presented in this and subsequent phases.

The raw data use four reserved numeric codes to mark non-substantive answers: -97 (“don’t know”), -98 (“not applicable”), -99 (“refusal”), and -999 (“irrelevant answer”, used only within the QK financial-knowledge block). As a first step, all four codes were recoded to missing values (NaN) across every variable except the survey weight.

For this project, the data preparation phase focuses on three groups of variables:

1. the *socio-demographic block* (gender, age, macro-region, type of municipality, education, employment status, household income bracket, nationality);
2. the *digital-behaviour blocks* (QD14, internet access; QP8 and QP9, digital financial activities);
3. the *financial-knowledge block* (QK3, QK5, QK6, QK7, QK10);

*A comprehensive dictionary mapping all the referenced survey codes to their verbatim questionnaire definitions is provided in **Appendix A**.*

2.2 Comments on strengths and weaknesses of the data

Strengths. The sample size ($N = 4,862$) is adequate for the descriptive and modelling steps planned in the following phases. The survey weight allows population-representative estimates. Internal consistency checks confirm that the conditional structure of the questionnaire is respected in the data: the 435 respondents with missing values on the QP8/QP9 digital-finance items correspond exactly to the 415 respondents without internet access plus 20 respondents who refused to answer the internet-access question (QD14), as expected from the questionnaire’s filter logic. No duplicate records were found.

Weaknesses. Two issues are worth flagging explicitly. First, the household income variable (QD13) has a very high non-response rate: 1,229 refusals and 284 “don’t know” answers, i.e. about 31% of the sample. Given the size of this group, dropping these respondents would discard nearly a third of the sample and could introduce bias (non-response on income is rarely random). We therefore retain these respondents and create an explicit “Not declared” category for income, rather than imputing or discarding them; this choice is revisited if income turns out to be an important predictor in the modelling phase. Second, 219 respondents (4.5%) refused to disclose their exact age (QD7) but did provide a 10-year age bracket (QD7_A); these respondents are kept in the analysis via a derived age-bracket variable, at the cost of not being usable in models that require a continuous age measure.

3 Methodology

3.1 The overall logic thread of the study

To achieve these preliminary goals and comprehensively answer the core research question, the study was structured into four interconnected sequential phases, as illustrated in Figure 1. This analytical pipeline maps the entire progression of the work, from raw data manipulation to the final interpretation of the results.

The logic thread begins with the first phase of data preparation, which establishes the foundational data architecture by pursuing three primary goals: (i) producing a clean, analysis-ready dataset with interpretable variable names and labels; (ii) operationalising the central concept — “non-adoption of digital financial tools” — into measurable variables; and (iii) defining “financial literacy” as a comparable score.

Once the dataset is consolidated, the second phase, concerning the descriptive analysis, conducts a profiling of non-adopters across socio-demographic dimensions. This lays the groundwork for the modelling phase, which executes the core multivariate analysis via weighted logistic regression and k -means clustering. Finally, the last phase of interpretation synthesises the empirical outputs to extract internally coherent vulnerability profiles.

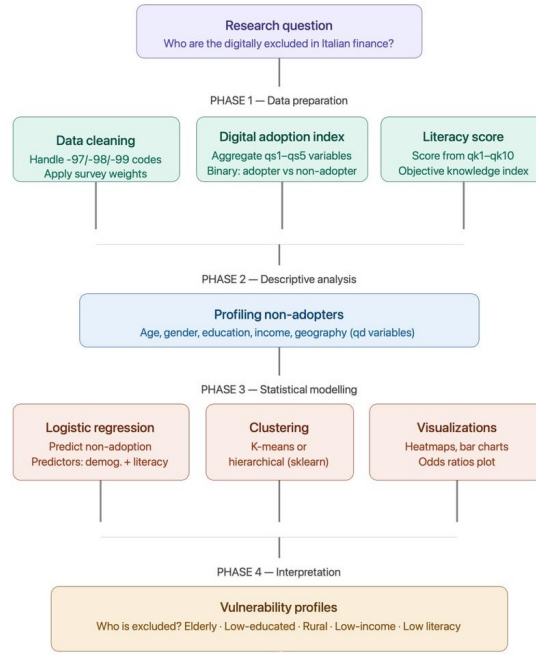


Figure 1: Analytical workflow and project pipeline

3.2 Digital adoption / digital exclusion.

Two distinct phenomena are distinguished, following the structure of the questionnaire itself:

- *Basic digital divide*: respondents without any internet access (QD14 = 0, $n = 415$, 8.5% of the sample) are, by construction, excluded from every digital financial service.
- *Digital financial exclusion among internet users*: among the 4,427 respondents with internet access, the QP8 block (5 binary items: “have you ever done X completely online”, e.g. opening an account, requesting a card, taking out insurance, a loan, or investing online) and the QP9 block (7 items on a 1-4 “never” to “very often” scale, e.g. checking balances, paying bills, transferring money, mobile payments, robo-advisors) are combined into a continuous **Digital Adoption Score (DAS)**, rescaled to 0-100. Internet users are then split into “low digital use” (DAS at or below the weighted 33rd percentile, threshold ≈ 13.9) and “digital adopters” (above the threshold).

These two pieces are combined into a single categorical variable, *digital_exclusion_status*, with three levels (“No internet access”, “Internet, low digital use”, “Internet, digital adopter”), plus a binary version (*digital_exclusion_binary*: 1 = excluded — no internet or low use — vs. 0 = adopter) for use in the logistic regression.

3.3 Financial literacy score

Following the standard OECD/INFE approach, an objective financial literacy score (0-7) is built from seven knowledge questions covering inflation (QK3), simple interest (QK5), compound interest (QK6), three true/false statements on risk-return, inflation and diversification (QK7_1/2/3), and a mortgage-related statement (QK10). Each item is coded 1 if answered correctly and 0 otherwise (including “don’t know” and missing answers, consistent with standard OECD/INFE scoring practice). The subjective self-assessment of financial knowledge (QK1) is kept as a separate variable and is not part of the objective score.

After the data preparation step, the second step of the analysis is a descriptive profiling of digital exclusion: for each of the socio-demographic dimensions identified as relevant in the research question (gender, age, education, geographic area, type of municipality, employment status, and household income), we compute the weighted distribution of the three-level *digital_exclusion_status* variable. This step answers, in descriptive terms, the question “who are the digitally excluded?”

and is a necessary precondition for the multivariate analysis: it identifies which variables show a strong association with digital exclusion (candidates for the logistic regression in the next step), and it also reveals where simple bivariate associations may be confounded by other variables, which directly motivates the move to a multivariate model.

The third phase addresses the core of the research question with two complementary statistical tools. The first is a weighted binary **logistic regression**, which estimates the independent association between each socio-demographic predictor and the binary digital-exclusion outcome (*digital_exclusion_binary*: 1 = excluded, 0 = adopter), controlling for all other predictors simultaneously. This directly addresses the confounding identified in the previous phase of Descriptive Analysis: for example, it allows us to ask whether high education is genuinely protective against exclusion once age is held constant. The second tool is a **k-means clustering analysis**, applied exclusively to the 4,427 internet users in the sample, which identifies distinct latent profiles of internet users who differ in their combination of digital adoption, financial literacy, age, income, and education.

The two approaches are intentionally complementary: the regression identifies *which variables matter* and in which direction, while the clustering identifies *who the people are* — a distinction that is particularly important for policy design, since interventions to reduce digital financial exclusion require targeting concrete population groups, not abstract regression coefficients.

3.4 The tools: which and why

The data preparation was carried out in *Python*, using *pandas* for data manipulation and *numpy* for numerical operations. These tools were chosen for their transparency (every recoding step is explicit and auditable) and because they integrate directly with the statistical and clustering libraries (*statsmodels*, *scikit-learn*) planned for the modelling phase.

For analysis, *matplotlib* is used to produce the descriptive charts (stacked bar charts of weighted percentages), in addition to *pandas/numpy* for the weighted cross-tabulations.

Logistic regression was chosen for three reasons: the outcome is binary with a moderate prevalence (43%), for which logistic regression has well-understood properties; the odds-ratio parameterisation is directly interpretable by the non-technical stakeholders (policy makers, financial educators) identified in the research question; it integrates naturally with survey weights through its maximum-likelihood estimator, preserving the representativeness of the IACOFI sample. Predictors include: gender (ref: Male), five age groups (ref: 30–39), five macro-regions (ref: North-West), education (ref: Medium), four employment groups (ref: Employed), four income brackets (ref: 1,751–2,900 EUR), two municipality-size groups (ref: Town 10k–50k), and the standardised financial literacy score. Rare employment categories (temporary employees, homemakers, unable to work) are merged into a “Student/Inactive” group to avoid sparse cells. The final regression sample is $N = 4,781$ after listwise deletion of the 81 cases with missing values on at least one predictor.

K-means clustering is applied on the subset of 3,051 internet users with complete data on the five clustering features: Digital Adoption Score (0–100), financial literacy score (0–7), imputed age, income (ordinal, 0–2), and education (ordinal, 0–2). The 415 respondents without internet access are intentionally excluded from this step: as a group they are homogeneous in their exclusion mechanism — absence of basic digital infrastructure — and their profile has already been characterised in the descriptive phase; the clustering is therefore focused on the more heterogeneous and more actionable group of internet users who nevertheless show low or absent digital financial adoption. Purely categorical variables (gender, region, employment) are intentionally excluded from the clustering feature space, because k-means relies on Euclidean distances that are not meaningful for unordered categories; these variables are used next to label the clusters once they have been identified. All five features are standardized (zero mean, unit variance) before clustering to prevent the wider-range variables (Digital Adoption Score, age) from dominating the distance metric. The number of clusters is selected by maximising the silhouette coefficient across $k = 2$ to 7. A PCA projection onto two dimensions is used for visual validation of cluster separation.

The Figure 2 presents a two-dimensional visual validation of the six-cluster solution using a Principal Component Analysis (PCA) projection. The horizontal axis represents the first principal component (PC1), which accounts for 31.2% of the total variance, while the vertical axis represents the second principal component (PC2), explaining an additional 22.1% of the variance. Together, these two components capture 53.3% of the variability in the standardized five-feature clustering space. Individual respondents are represented as data points, color-coded according to their as-

signed cluster (Clusters 0 through 5). The ‘X’ marks highlight the respective cluster centroids. The plot demonstrates the spatial distribution and relative overlap of the segments: while some clusters (such as Cluster 2 on the left and Cluster 5 on the far right) exhibit clearer spatial segregation along PC1, the central regions show a higher density of overlapping points, reflecting the continuous nature of the underlying behavioral and demographic features among internet users.

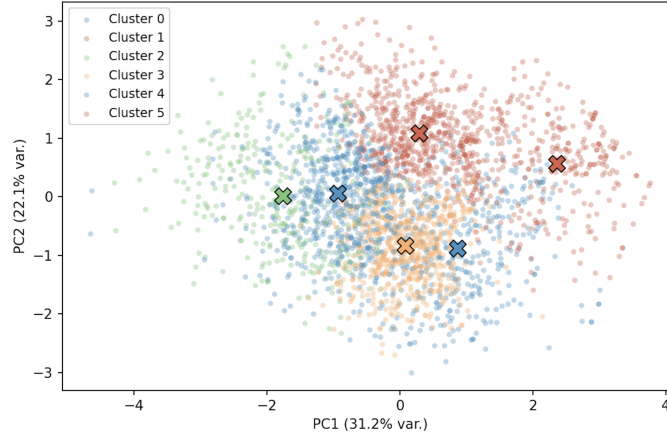


Figure 2: PCA projection of clustering solution

4 Results

4.1 The steps of the study

The cleaned dataset (*cleaned_dataset.csv*, 4,862 rows $\times$ 16 columns) contains the following derived variables, ready for the descriptive profiling of the second phase:

Table 1: Derived variables in the cleaned dataset.

Variable	Description	Missing
<code>digital_exclusion_status</code>	3-level: No internet / Internet+low use / Internet+adopter	1.3%
<code>digital_exclusion_binary</code>	1 = excluded, 0 = adopter	1.3%
<code>digital_adoption_score</code>	0–100, internet users only	9.8%
<code>financial_literacy_score</code>	0–7, objective score	0%
<code>income_bracket</code>	Incl. explicit “Not declared” category	0%
<code>age / age_bracket</code>	Continuous age (4.5% missing) / 10-year bracket (0% missing)	—

The weighted distribution of the digital exclusion variable is:

- No internet access: 10.7%
- Internet, low digital use: 32.3%
- Internet, digital adopter: 57.0%

As a first check on the relationship between the two constructed indices, the weighted mean financial literacy score (overall: 3.61 out of 7) decreases monotonically across the three digital-exclusion groups: 4.05 among digital adopters, 3.35 among internet users with low digital use, and 2.42 among respondents with no internet access. This pattern is consistent with the research question's premise and motivates the descriptive and modelling work of the subsequent phases.

4.1.1 Descriptive profiling of digital exclusion

Overall picture. At the population level, 10.7% of respondents have no internet access at all, a further 32.3% have internet access but make low use of digital financial tools, and 57.0% are “digital adopters” — for an overall **digital exclusion rate of 43.0%** (sum of the first two groups).

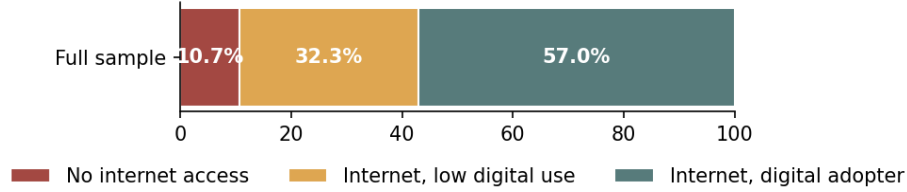


Figure 3: Overall digital exclusion status (weighted %)

By gender. Males show a higher exclusion rate (48.8%) than females (36.9%).

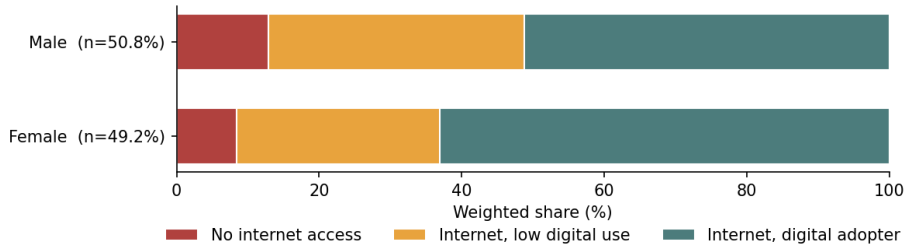


Figure 4: Digital exclusion status by gender (weighted %)

By age group. The relationship between age and digital exclusion shows that exclusion is lowest among 30-49 year-olds (23.5-23.9%), rises moderately for 50-59 year-olds (34.0%), and becomes very high for 60-69 (60.1%) and 70-79 year-olds (76.1%). Strikingly, the youngest group (18-29) also shows a high exclusion rate (53.9%), almost entirely driven by the “low digital use” category rather than lack of internet access.

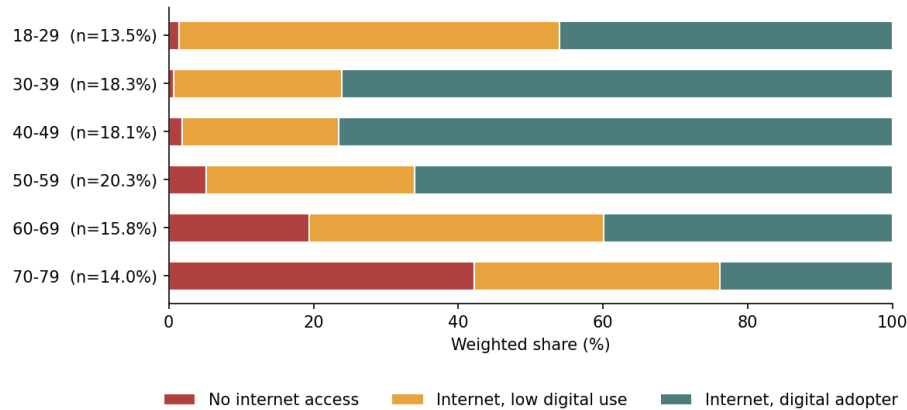


Figure 5: Digital exclusion status by age group (weighted %)

By education. The raw association with education is counter-intuitive: exclusion is lowest for “Low” education (18.0%), intermediate for “Medium” (33.9%), and highest for “High” education (75.9%).

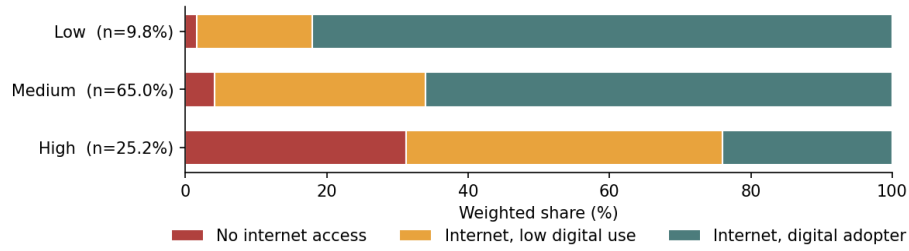


Figure 6: Digital exclusion status by education level (weighted %)

By geographic area. Differences across macro-regions are modest: exclusion ranges from 39.1% (Centre) to 47.5% (South), with North-West, North-East and the Islands all close to 40-44%.

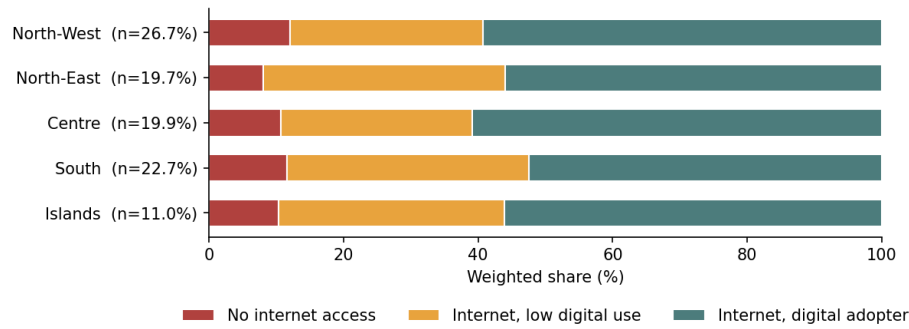


Figure 7: Digital exclusion status by geographic area (weighted %)

By type of municipality. The rural/urban gradient is present but small: 44.6% in rural municipalities (under 10,000 inhabitants) vs. 38.9% in urban areas (over 50,000 inhabitants), with towns in between (43.7%).

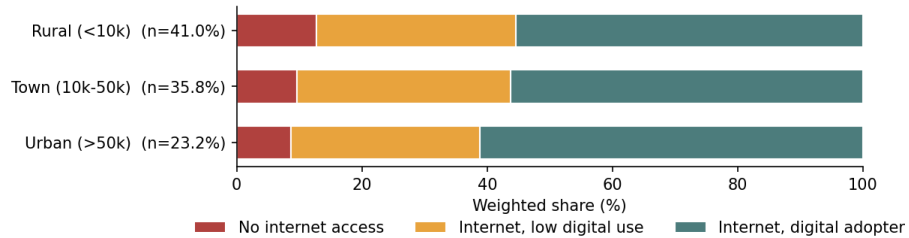


Figure 8: Digital exclusion status by type of municipality (weighted %)

By employment status. Employed respondents have by far the lowest exclusion rate (23.7%), while retirees (71.7%), students (69.3%) and people not in the labour force (66.2%) all show high exclusion — though, as discussed below, the reasons are likely very different across these groups.

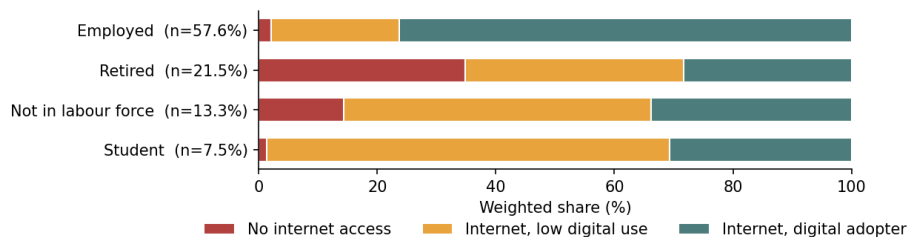


Figure 9: Digital exclusion status by employment status (weighted %)

By household income. Exclusion decreases monotonically with declared income: 50.7% for the lowest bracket (up to €1,750), 29.0% for the middle bracket, and 18.0% for the highest bracket

(over €2,900). Respondents who did not declare their income show a high exclusion rate (53.3%), similar to the lowest income bracket — consistent with the idea that income non-disclosure is itself correlated with broader forms of exclusion, though this remains speculative.

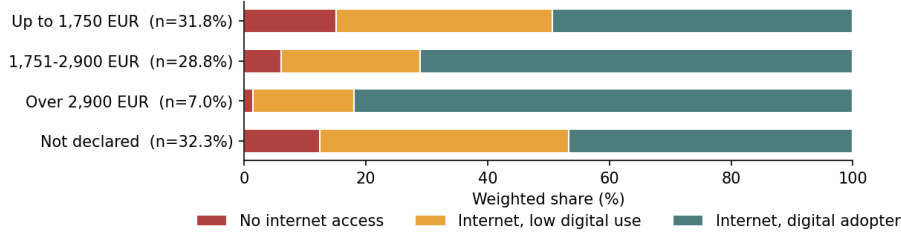


Figure 10: Digital exclusion status by household income (weighted %)

4.1.2 Financial literacy by education and exclusion status

Within every education level, the weighted mean financial literacy score increases monotonically from “No internet access” to “Internet, low digital use” to “Internet, digital adopter” (e.g., for “Medium” education: $2.92 \rightarrow 3.43 \rightarrow 4.06$). However, *across* education levels the pattern is again counter-intuitive: “Low” education shows higher literacy scores (3.38-4.54) than “High” education (2.23-3.35) in every exclusion category.

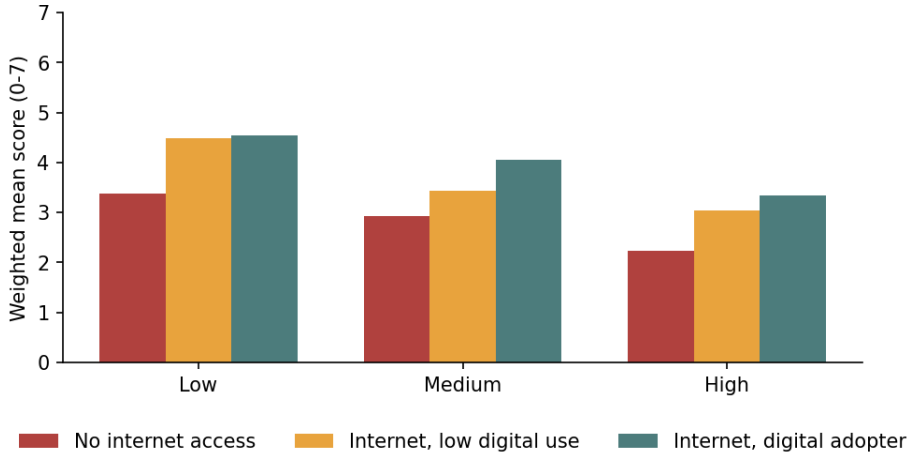


Figure 11: Financial literacy score by education level and digital exclusion status

4.1.3 Weighted logistic regression

The model fits well: $R^2 = 0.228$, with a log-likelihood ratio test significant at $p < 0.001$ ($-2LL$ improvement from null: 1,451 points on 20 degrees of freedom). The odds ratios with 95% confidence intervals are displayed in Figure 12. The key findings are as follows.

Gender. Being female is associated with a significantly *lower* odds of digital financial exclusion ($OR = 0.68$, 95% CI: 0.59–0.78, $p < 0.001$). This reverses the raw descriptive finding of previous phase, where females appeared less excluded than males (36.9% vs. 48.8%): after controlling for age, income, employment and education, women are *less* likely to be digitally excluded than men, not more.

Age. Compared to the 30–39 reference group, exclusion odds are significantly higher for adults aged 60+ ($OR = 2.25$, CI: 1.70–2.97) and 50–59 ($OR = 1.32$, CI: 1.04–1.67), and also for the 18–29 age group ($OR = 1.67$, CI: 1.27–2.19). Adults aged 40–49 do not differ significantly from the 30–39 reference group.

Education. The education effect is entirely reversed under multivariate control: low education is no longer significantly associated with exclusion once age is held constant ($OR = 0.82$, $p = 0.13$), while *high* education is strongly associated with *greater* exclusion ($OR = 3.70$, CI: 2.97–4.62, $p < 0.001$). This is the most striking finding of the regression and warrants careful interpretation, that is carried after.

Employment. Retired respondents (OR = 2.94, CI: 2.27–3.79) and the student/inactive group (OR = 3.70, CI: 3.00–4.56) are both strongly more likely to be excluded relative to employed respondents. Self-employed respondents do not differ significantly from employees.

Income. The income gradient is clear and in the expected direction: the lowest bracket (up to €1,750 per month) is associated with 78% higher odds of exclusion relative to the middle bracket (OR = 1.78, CI: 1.48–2.13), and the highest bracket (above €2,900) is associated with 28% lower odds (OR = 0.72, CI: 0.53–0.98). Respondents who declined to report their income also show elevated exclusion odds (OR = 1.82, CI: 1.52–2.18), consistent with income non-disclosure being correlated with marginalisation.

Geography. Living in the South is associated with modestly higher exclusion odds (OR = 1.28, CI: 1.05–1.56, $p = 0.014$) relative to the North-West. Other regions do not differ significantly. Urban respondents (cities above 50,000 inhabitants) show lower exclusion odds than residents of medium-sized towns (OR = 0.78, CI: 0.64–0.94, $p = 0.008$).

Financial literacy. Each one-standard-deviation increase in the objective financial literacy score is associated with a 18% reduction in the odds of digital exclusion (OR = 0.82, CI: 0.76–0.88, $p < 0.001$), independently of all socio-demographic controls. This is one of the most policy-relevant findings: even after adjusting for age, income and education, financial knowledge is independently protective against digital financial exclusion.

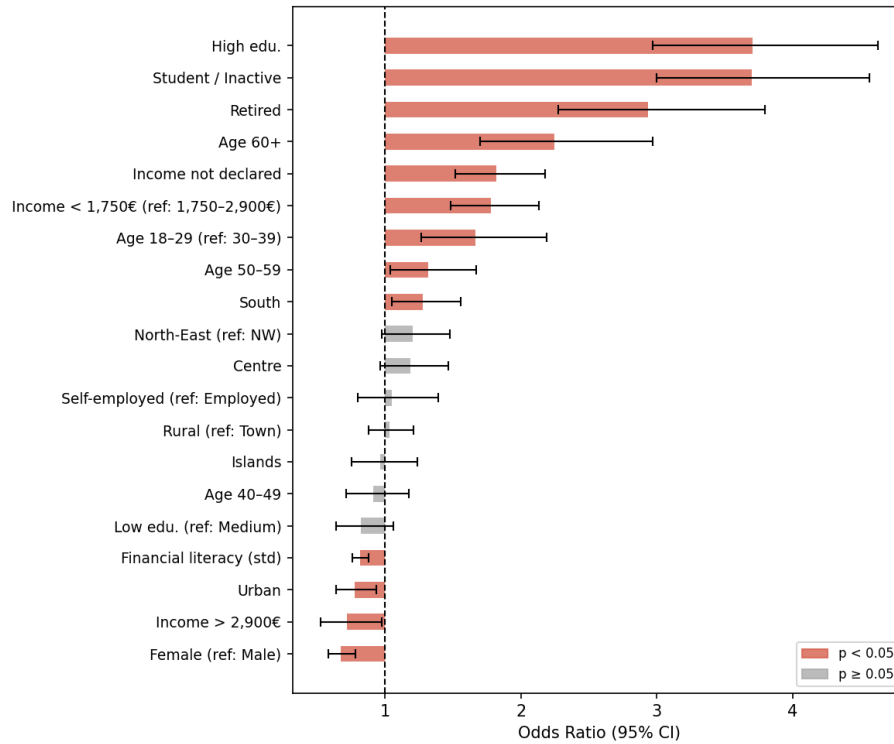


Figure 12: Forest plot of odds ratios

Table 2: Weighted logistic regression: predictors of digital financial exclusion

Predictor	Coef.	OR	95% CI	p-value
<i>Gender</i> (ref. Male)				
Female	−0.391	0.68	[0.59, 0.78]	< 0.001
<i>Age group</i> (ref. 30–39)				
18–29	0.510	1.67	[1.27, 2.19]	< 0.001
40–49	−0.090	0.91	[0.71, 1.17]	0.479
50–59	0.275	1.32	[1.04, 1.67]	0.024
60+	0.809	2.25	[1.70, 2.97]	< 0.001
<i>Region</i> (ref. North-West)				
North-East	0.184	1.20	[0.98, 1.48]	0.080
Centre	0.173	1.19	[0.97, 1.47]	0.104
South	0.247	1.28	[1.05, 1.56]	0.014
Islands	−0.036	0.96	[0.75, 1.24]	0.778
<i>Education</i> (ref. Medium)				
Low	−0.194	0.82	[0.64, 1.06]	0.132
High	1.310	3.70	[2.97, 4.62]	< 0.001
<i>Employment</i> (ref. Employed)				
Self-employed	0.052	1.05	[0.80, 1.39]	0.717
Retired	1.077	2.94	[2.27, 3.79]	< 0.001
Student/Inactive	1.308	3.70	[3.00, 4.56]	< 0.001
<i>Income bracket</i> (ref. 1,751–2,900 EUR)				
Up to 1,750 EUR	0.576	1.78	[1.48, 2.13]	< 0.001
Over 2,900 EUR	−0.332	0.72	[0.53, 0.98]	0.036
Not declared	0.598	1.82	[1.52, 2.18]	< 0.001
<i>Municipality size</i> (ref. Town 10k–50k)				
Rural (<10k)	0.031	1.03	[0.88, 1.21]	0.701
Urban (>50k)	−0.254	0.78	[0.64, 0.94]	0.008
Financial literacy score (std.)	−0.202	0.82	[0.76, 0.88]	< 0.001
Constant	−1.789	0.17	[0.13, 0.22]	< 0.001

Notes: $N = 4,781$. Outcome: digital_exclusion_binary (1 = excluded, 0 = adopter). McFadden pseudo- $R^2 = 0.228$. Log-likelihood: −2461.5 (null: −3187.0). AIC = 4965.1; BIC = 5101.0. OR = odds ratio; CI = confidence interval. Reference categories noted in italics above each block.

4.1.4 K-means clustering

Choosing the number of clusters k is not a purely statistical decision, since k -means requires k to be fixed in advance and will always partition the data regardless of whether genuine sub-groups exist. Two complementary diagnostics are therefore combined. The elbow method (Figure 13 on the left) tracks the within-cluster sum of squares (inertia) as k increases: inertia decreases monotonically by construction, so the goal is not to find a minimum but to identify the point beyond which adding clusters yields diminishing returns. The silhouette coefficient (Figure 13 on the right) complements this by measuring, for each point, how much closer it is to its own cluster than to the nearest neighbouring cluster, averaged across all points; values closer to 1 indicate well-separated, compact clusters, while values near 0 indicate overlapping clusters with no clear boundary. In our case, the inertia curve in Figure 13 does not show a sharp elbow, but flattens progressively from $k=4$ onward, suggesting a plausible range of 4 to 6 clusters — an expected pattern given that the underlying population is genuinely composed of overlapping, continuous gradients (age, income, literacy) rather than naturally separated groups. The silhouette scores are uniformly modest (0.18–0.22), which is typical for survey data of this kind and reflects the fact that demographic and behavioural variables vary continuously across the population rather than forming tight, well-isolated clusters; nevertheless, the score is maximised at $k=6$ (0.216). We therefore retain $k=6$ as the working solution, while explicitly noting that the silhouette values indicate moderate rather than strong cluster separation, a limitation discussed further.

Table 3: Cluster centroids and exclusion rates (internet users, N = 3,051)

Cluster	n	DAS	Fin. Lit.	Age	Inc.	Edu.	% Excl.	Label
0	401	19.2	1.6	47	0.44	1.00	43%	<i>Doubly vulnerable</i>
1	684	18.5	5.0	64	0.91	1.00	41%	<i>Literate late adopters</i>
2	346	34.0	4.9	47	1.05	0.00	14%	<i>Low-edu practical adopters</i>
3	710	24.2	5.0	36	0.26	1.02	26%	<i>Young constrained</i>
4	649	48.4	4.3	47	1.20	1.03	0%	<i>Full digital adopters</i>
5	261	11.2	3.7	64	0.30	2.00	72%	<i>Educated but excluded elderly</i>

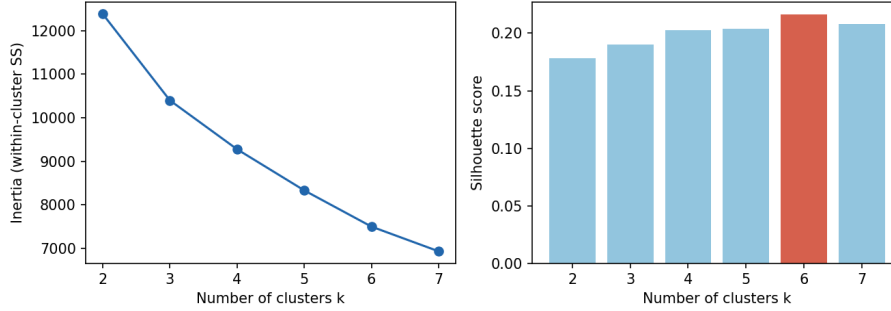


Figure 13: Elbow method and Silhouette coefficients

The Cluster profiles (Figure 14) displays the six cluster centroids on a radar chart, with each of the five clustering features (Digital Adoption, Financial Literacy, Age, Income, Education) normalised to a common 0–1 scale so that clusters with very different absolute values can be compared on the same axes; a centroid that bulges outward on a given axis indicates that the cluster scores high on that feature relative to the other clusters, while a centroid pulled toward the centre indicates a low relative score. This visualisation makes the heterogeneity documented in Table 3 immediately legible: Cluster 4 (Full digital adopters) is pulled sharply toward the Digital Adoption and Income vertices while remaining close to the centre on Age, describing a profile of younger, higher-income, highly engaged users. Cluster 5 (Educated but excluded elderly) shows the mirror-image pattern — pulled toward Age and Education but collapsed toward the centre on Digital Adoption — visually confirming that this group’s exclusion is not explained by a lack of formal education or by youth-related unfamiliarity with technology, but by some other behavioural or attitudinal barrier not captured in the five clustering features. Clusters 2 and 3 occupy an intermediate position, both scoring near the maximum on Financial Literacy while differing sharply on Income and Age, illustrating that financial knowledge alone does not determine which side of the digital divide a respondent falls on. The radar chart therefore provides a visual confirmation of the regression findings: digital adoption and financial literacy are related but partially independent dimensions, and the population separates along both simultaneously rather than along a single excluded-versus-included axis.

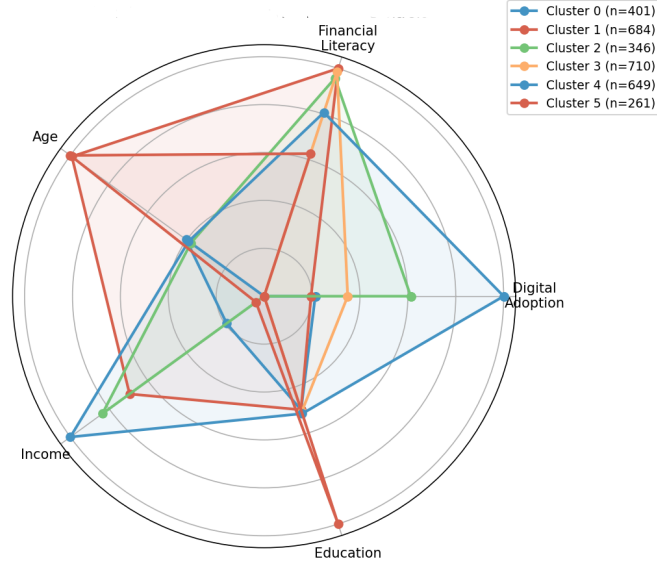


Figure 14: Cluster profiles

The distribution of key variables by cluster in Figure 15 complements Table 3 by showing the full distribution of the three most informative clustering variables across clusters, using box plots. In a box plot, the box spans the interquartile range (the middle 50% of observations, from the 25th to the 75th percentile), the horizontal line inside the box marks the median, the whiskers extend to the most extreme values within 1.5 times the interquartile range, and individual points beyond the whiskers are plotted as outliers; this representation is preferable to bar charts of means alone because it reveals within-cluster heterogeneity and skewness that a single centroid value would conceal. Three patterns are worth highlighting:

- the *Digital Adoption Score* panel shows that Cluster 4 (Full digital adopters) and Cluster 5 (Educated but excluded elderly) occupy almost completely non-overlapping ranges — Cluster 4’s interquartile range sits at 39–56, while Cluster 5’s sits at 0–17 — confirming that the separation between these two groups on the digital-adoption axis is not an artefact of the centroid alone but holds for the large majority of individuals within each group
- the *Financial Literacy* panel shows that Clusters 1, 2 and 3 have nearly identical, high distributions (median = 5 out of 7, with overlapping interquartile ranges), which visually confirms the radar-chart finding that financial literacy alone cannot distinguish these clusters; what separates them is instead age and income, as shown in the third panel.
- the *Age* panel shows that Clusters 1 and 5 — the two profiles with the highest exclusion rates (41% and 72%) — are also the only two clusters whose interquartile range lies entirely above age 58, with median ages of 65 and 67 respectively; the box plot makes clear that this is not driven by a handful of elderly outliers but reflects the typical respondent in both groups.

Taken together, those boxplots reinforces the conclusion that age is the dimension most consistently associated with severe digital exclusion among internet users, while financial literacy is unevenly distributed across age groups and therefore only partially able to offset the age-related disadvantage.

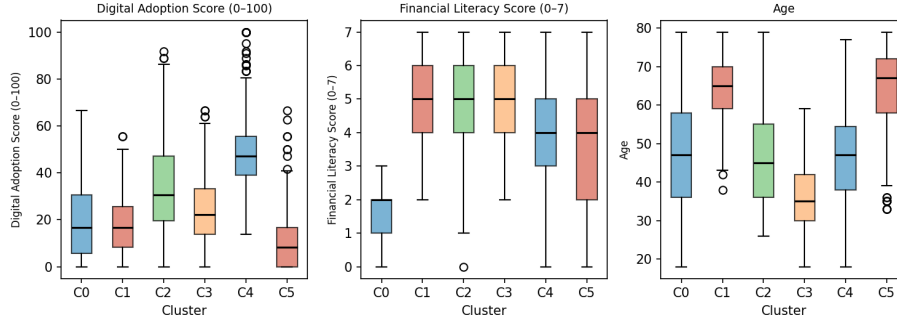


Figure 15: Distribution of key variables by cluster

4.2 Comments and interpretation

4.2.1 Interpreting descriptive profile

The descriptive results combine two different stories and distinguishing them is itself one of the findings of this phase.

An expected “digital divide” pattern. Several results align with the classic digital-divide narrative that motivated the research question: exclusion is strongly associated with *older age* (60+), *lower household income* and *rural municipalities*. These results are consistent with prior literature on digital and financial exclusion and support the relevance of the research question for this population.

A confounded, counter-intuitive pattern for education (and youngest age group).

At first, the raw cross-tabulations are puzzling: “High education” respondents show a much higher exclusion rate (75.9%) than “Low education” respondents (18.0%), and the youngest age group (18–29, 53.9%) is also highly excluded. Both patterns run against intuition — and both turn out to be driven by the same variable: age.

The supporting diagnostic in Figure 16 shows that the “High education” group in this sample is heavily skewed toward older cohorts: 57.8% of “High education” respondents are aged 60 or above, compared to just 12.0% of “Low education” respondents. This reflects a well-known generational pattern in Italy, where university attendance was far less common several decades ago; as a result, today’s older university-educated respondents represent a historically small and specific group, not a typical highly-educated profile.

Crucially, both the Digital Adoption Score and the financial literacy score follow a U-shaped pattern by age: lowest among the youngest (18–29) and oldest (70–79) respondents, and highest among 30–59 year-olds. This means the education cross-tabs are not really capturing an education effect at all — they are capturing this same age-related U-shape, simply relabelled through the lens of education. In other words, **education looks associated with exclusion only because, in this sample, “highly educated” disproportionately means “elderly”, and elderly respondents are independently more exclusion-prone.** The same logic explains the youngest group’s high exclusion rate: 18–29-year-olds sit at the other end of the same U-shape, not because of low education, but because of their position in the same age-driven curve.

A similar life-stage explanation likely applies to the high exclusion rate among **students** (69.3%): the Digital Adoption Score is built in part from QP8 items (“have you ever done X completely online”, e.g. opened an investment account, taken out a loan), which mechanically score low for anyone who has simply never engaged in these financial activities yet — regardless of their digital skills. Students and very young adults are disproportionately affected by this measurement issue.

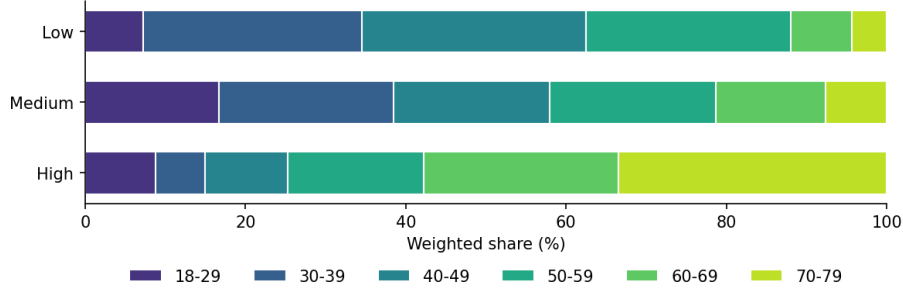


Figure 16: Age composition within each education level (weighted%)

4.2.2 Implication

These findings validate the digital adoption / exclusion index, but they show that univariate cross-tabs are not sufficient to identify which socio-demographic factors are independently associated with digital exclusion. Age, education, employment status and income are correlated with each other in ways that can mask their individual associations with exclusion. This is precisely the motivation for the multivariate logistic regression, which estimate the association between each socio-demographic variable and digital exclusion while holding the others constant. The gender gap (48.8% vs. 36.9%) is a further candidate for this kind of multivariate check, since it is not obviously explained by age composition alone (see Figure 4 the male/female age distribution).

4.2.3 Interpreting the modelling phase

The education paradox The most striking regression result — that high education is associated with dramatically higher exclusion odds ($OR = 3.70$) even after controlling for age — is not immediately intuitive. It is important to note, however, that the “High” education group in this sample is almost entirely composed of older adults (Cluster 5: avg. age 64, 71% with secondary-plus education) who have internet access but do not use digital financial tools. This group has above-median financial literacy but very low Digital Adoption Scores, and is distinguished from the rest of the sample by a behavioural pattern that is consistent with non-adoption: they have the cognitive resources to adopt but choose not to, possibly due to trust barriers, habit inertia, or a preference for face-to-face banking interactions. The regression coefficient is therefore best interpreted as a group effect operating through employment and life stage, not as “education causes exclusion.”

Financial literacy as independent protective factor. The independent, statistically significant negative association between financial literacy and digital exclusion is arguably the study’s most actionable finding. It suggests that interventions targeting objective financial knowledge — not just digital skills — could reduce digital financial exclusion, even among groups that are constrained by age or income. This result survives full socio-demographic control, strengthening the causal interpretation.

Heterogeneity within the excluded population. The clustering results reveal that the digitally excluded are not a homogeneous group. Among internet users, three qualitatively different exclusion profiles emerge: (i) *Doubly vulnerable* (Cluster 0, 43% excluded): middle-aged adults with simultaneously very low financial literacy and very low income, for whom both knowledge and financial constraints may be barriers; (ii) *Literate late adopters* (Cluster 1, 41% excluded): older adults with high financial literacy who nevertheless show low digital adoption — a profile consistent with behavioural barriers (trust, habits) rather than knowledge deficits; (iii) *Educated but excluded elderly* (Cluster 5, 72% excluded): the highest-exclusion group — old, highly educated, low income, very low DAS — likely representing university-educated retirees who have internet access but have never engaged with digital financial services. These three profiles require very different policy responses: financial education for Cluster 0, trust-building for Cluster 1, and possibly peer-mediated digital onboarding for Cluster 5.

Gender reversal under multivariate control. The finding that females are significantly *less* likely to be excluded after adjustment ($OR = 0.68$) is a notable instance of Simpson’s paradox: in the raw data, females appeared more excluded (36.9% vs. 48.8%), but this was explained by the fact that excluded males in this sample are disproportionately employed in lower-income brackets and more likely to be young. Once these factors are held constant, women are actually more

inclined to adopt digital financial tools — a finding that challenges the conventional narrative of female digital financial disadvantage in Italy.

5 Conclusions

5.1 Gap between the goal and the results

The research question asked: *what socio-demographic and financial literacy characteristics predict non-adoption of digital financial tools among Italian adults, and do identifiable vulnerability profiles exist?* The study provides a substantive, if not complete, answer to both parts.

On the first part — **predictors of exclusion** — the goal is substantially met. The logistic regression identifies a clear and interpretable set of independent risk factors: older age (especially 60+), low household income, retirement or inactivity, residence in the South or in non-urban municipalities, and lower financial literacy. The gender result is more nuanced than anticipated: raw data suggested males were more excluded, but multivariate analysis reveals a Simpson’s paradox, with females actually less at risk once other factors are controlled for. The education finding — that high formal education is associated with greater exclusion — is counterintuitive but explained by cohort composition: it reflects an age effect specific to the Italian educational history, not a genuine causal relationship between university-level credentials and digital financial disengagement.

On the second part — **vulnerability profiles** — the k-means clustering answers the question with considerable granularity. Among internet users, three distinct excluded profiles are identified. The *Doubly vulnerable* group (Cluster 0) combines low financial literacy with low income and moderate digital adoption, suggesting that both knowledge and resource constraints operate simultaneously. The *Literate late adopters* (Cluster 1) are older adults with high financial knowledge but low digital adoption, pointing to behavioural rather than cognitive barriers. The *Educated but excluded elderly* (Cluster 5) represent the most severely excluded group (72%), characterised by high education, high age, and very low income — a profile consistent with university-educated retirees who have chosen not to engage with digital financial services. In addition, a fourth group worth monitoring is the *Young constrained* cluster (Cluster 3, 26% excluded): young adults with high financial literacy but very low income who are partially excluded, likely because financial constraints — rather than knowledge or attitude — prevent engagement with digital financial products.

Where the study falls short of its original goal is on the causal dimension. The cross-sectional design of IACOFI prevents any causal inference: we cannot determine, for instance, whether low financial literacy leads to non-adoption of digital tools or whether non-adoption reduces the opportunities for building financial knowledge. Similarly, the study does not capture *why* individuals in the excluded clusters refrain from digital financial adoption: trust, habit, usability barriers, privacy concerns, or simply satisfaction with existing channels are all plausible mechanisms that the current data cannot distinguish. These are gaps between the stated goal and what the results can deliver.

5.2 Strengths and weaknesses of the study

Strengths. The principal strength of this study is the quality and representativeness of the underlying data. IACOFI 2023 is a large ($N = 4,862$), nationally representative, probability-sampled survey conducted by the Banca d’Italia, with survey weights calibrated to the Italian adult population. The 2023 edition is also the first wave to include digital financial competence items following the OECD/INFE methodology, making it uniquely suited to the research question. The two-level operationalisation of digital exclusion — distinguishing those without internet access from those with access but low digital financial use — is a methodological contribution. The combination of logistic regression and clustering is analytically coherent: the former answers the “which predictors matter” question with statistical rigour, the latter answers the “who are the excluded” question with population-level granularity. The identification and separate treatment of confounding is a further analytical strength.

Weaknesses. Four weaknesses are worth flagging. First, the cross-sectional design rules out causal inference and prevents the study from tracking how digital adoption evolves over time for the same individuals. Second, the **Digital Adoption Score** is built from QP8/QP9 items that conflate “never done because I choose not to” with “never done because the financial product is not

relevant to my life stage” — a measurement ambiguity that particularly affects young adults and students, who score low on DAS simply because they have not yet opened investment accounts or taken out loans, regardless of their digital capability. Third, as noted in Section 2.2, the income variable’s 31% non-response rate — addressed via a “Not declared” category — introduces a degree of imprecision in the income coefficient estimated in the regression. Fourth, the k-means algorithm operates on Euclidean distances and is sensitive to the initial centroid placement; although the procedure was repeated with 30 random initialisations and validated with the silhouette coefficient, the cluster solution is not guaranteed to be globally optimal, and a different algorithm might yield a modestly different partition.

5.3 Directions for further research and improvements

Four directions stand out as most productive for future work.

Longitudinal design. The most impactful improvement would be a design that follows the same individuals across IACOFI waves (2020 and 2023 are available). This would allow researchers to identify whether financial education interventions between waves actually reduce digital exclusion in subsequent observations, and to model the life-course trajectory of digital financial adoption — addressing the central causal gap of the current study.

Barrier-specific measurement. A targeted qualitative or mixed-methods study on the excluded clusters — particularly the *Literate late adopters* and *Educated but excluded elderly* — would be invaluable for distinguishing between trust-based, habit-based, and usability-based barriers to digital financial adoption. The IACOFI questionnaire contains attitudinal items in the QS4 block (trust in online banks and FinTechs) that were not used in the current analysis and could serve as a starting point for such an extension within the same dataset.

Improved DAS construction. Future versions of this study should address the life-stage ambiguity in the Digital Adoption Score by constructing age-stratified versions of the index, or by weighting QP8 items by their relevance to each respondent’s financial situation (e.g. excluding investment-platform items for respondents without savings). This would produce a fairer measure of digital capability rather than digital activity, particularly for young adults and students.

Policy targeting and evaluation. The cluster profiles identified can be used directly as targeting criteria for financial inclusion interventions. The Banca d’Italia’s ongoing financial education programmes could be designed with cluster membership in mind: literacy-focused programmes for Cluster 0, digital trust-building campaigns for Clusters 1 and 5, and income-support or fee-waiver schemes for Cluster 3. An experimental evaluation of such targeted interventions, using future IACOFI waves as follow-up, would close the causal gap this study has identified and provide evidence on what actually works to reduce digital financial exclusion in Italy.

Overall, this study demonstrates that digital financial exclusion in Italy is a structured, heterogeneous phenomenon driven by an identifiable combination of age, income, employment status, and financial knowledge. The 43% overall exclusion rate, with meaningfully different profiles underneath it, points to the need for differentiated policy responses rather than a single universal intervention. The finding that financial literacy is independently protective, above and beyond socio-demographic factors, reinforces the case for continued investment in financial education as a lever for digital financial inclusion — one of the central policy commitments of the Banca d’Italia’s strategic agenda for the coming years [2].

Appendix A Survey Variable Dictionary

Variable	IACOFI 2023 Questionnaire Item / Construct Verbatim Description
Socio-Demographic Predictors (QD Block)	
QD1	Gender of the respondent (1 = Male, 0 = Female)
QD2	Macro-region of residence (1 = North-West, 2 = North-East, 3 = Centre, 4 = South, 5 = Islands)
QD3	Community size (1 = Rural < 3k, 2 = Small town 3k–15k, 3 = Town 15k–100k, 4 = City 100k–1M, 5 = Large city > 1M)
QD7	Age of the respondent in completed years (Minimum value = 18)
QD7_A	Age band in 10-year intervals (recorded for respondents refusing exact age QD7)
QD9	Highest level of education completed (10-category scale collapsed to: Low, Medium, High)
QD10	Main working status (10-category scale collapsed to: Employed, Retired, Student, Inactive)
QD13	Net monthly household income bracket (1 = Up to 1,750 EUR, 2 = 1,751–2,900 EUR, 3 = Over 2,900 EUR; includes derived “Not declared” category for codes -97/-99)
Digital Access & Financial Behaviour (QD / QP Blocks)	
QD14	Basic digital divide filter: access to the internet (1 = Yes, 0 = No)
QP8	Digital adoption binary block (1 = Yes, completely online): _1 Opened a current/payment or savings/deposit account _2 Requested a payment, debit, or credit card _3 Subscribed to an insurance policy _4 Taken out credit/loan _5 Invested money via online platforms (crowdfunding, crowd investing, peer-to-peer lending)
QP9	Digital usage frequency over the past 12 months (Scale: 1 = Never, 4 = Very often): _1 Checked balance and transactions online _3 Paid bills online _4 Bought goods and services online _5 Transferred money to others online _6 Managed financial products and services online _7 Paid in a physical shop with a mobile phone (mobile wallet) _10 Consulted an online platform for automated financial advice (roboadvisor)
Financial Literacy Block (QK Block)	
QK1	Subjective literacy: self-rated overall financial knowledge compared with others (Scale: 1 = Very high, 5 = Very low)
QK3	Objective literacy (Item 1): purchasing power of 1,000 EUR in one year with inflation at 8%
QK5	Objective literacy (Item 2): calculation of simple interest (100 EUR at 2% per year after 1 year)
QK6	Objective literacy (Item 3): calculation of compound interest (100 EUR at 2% per year after 5 years)
QK7_1	Objective literacy (Item 4): True/False — “An investment with a high return is likely to be high risk”
QK7_2	Objective literacy (Item 5): True/False — “High inflation means that the cost of living is increasing rapidly”
QK7_3	Objective literacy (Item 6): True/False — “It is usually possible to reduce stock market risk by buying a wide range of stocks”
QK10	Objective literacy (Item 7): True/False — “A 15-year mortgage has higher monthly payments than a 30-year mortgage, but total interest paid is lower”
Survey Mechanics	
wght	Individual survey weight calibrated to restore representativeness of the Italian adult population

Table 4: Verbatim dictionary of the IACOFI 2023 survey variables.

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